

Virtual Robot Riding with a Person in a Real Elevator - Mixed Reality System for Robot Behavior Design -*

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Abstract— To provide more services on multiple floors in buildings, recent mobile robots have the capability to take elevators and need to board them with other passengers. When a mobile robot boards an elevator, the current technological limitations result in stress for the surrounding passengers. However, it is still unclear what impressions a robot's quick boarding behavior will impart to surrounding passengers, when such limitations are removed in the future and mobile robots acquire human-like quick movements. It is also difficult to increase the speed of robots drastically in a manner which ensures the safety of experiment participants. In this study, we developed a mixed reality system that integrates a virtual robot into the real world to ensure the physical safety of participants during the experiment. Visual and auditory information about the virtual robot was provided to participants through a head-mounted display and was synchronized with the robot's behavior. We focused on three basic factors of robot's boarding behavior: forward speed during entry, rotation speed, and rotation direction in the elevator car. A statistical analysis revealed that quick, human-like entry into an elevator by a virtual robot reduces perceived anthropomorphism, likeability, intelligence, and safety, and increases the stress levels experienced by the passengers. The statistical analysis also showed that quick rotation inside the elevator by the virtual robot reduces perceived likeability. Our results provide useful knowledge for designing the behavior of future robots when they board elevators with other passengers.

I. INTRODUCTION

In recent years, more service robots have been taking elevators to provide more services on multiple floors of buildings [1]. Many studies have focused on the identification [2-5] and the operation of the control panels of elevators [5-7]. In the current collaborations between robots and elevators, there is a need to transport both people and a service robot simultaneously in elevator cars to improve the efficiency of elevator transport. More studies thus have focused on the behavioral design of service robots that perform more socially acceptable behaviors when using elevators with people, such as waiting for an elevator or actually riding in an elevator [8-11].

Although today's service robots need to improve their mobility to provide higher-quality services, they cannot move as fast as humans for multiple reasons, such as hardware

limitations, communication delays, the need to ensure the safety of humans around them. When taking an elevator, a service robot needs to move slowly to ensure a solid connection with the elevator system and to minimize the impact on passengers in the elevator car and along with people in the elevator hall. As a result, a robot cannot board an elevator as smoothly as a human, which causes stress for the surrounding passengers who have to wait for the robot's slow boarding. [12,13].

In the future, these technical challenges will be solved, and service robots will achieve a high level of mobility that will allow them to board elevators as smoothly as humans. On the other hand, passengers try to keep their distance from service robots in elevator cars due to the limited space available inside [14]. Therefore, even if a service robot is safe from a technical standpoint, passengers will still feel psychological stress toward the robot because it is physically impossible to maintain a sufficient distance within the confined space of the elevator car. It remains largely unclear how the future improved robot mobility and boarding behavior will affect surrounding passengers.

Some studies have begun to address this problem. For example, [15] studied the effect of robot's boarding speed on people and found that fast robot's boarding speed contributed to enhance people's satisfaction during boarding. However, in [15], an online comparative survey using video-based scenarios was conducted which gave the participants different impressions from those of actual elevator passengers. Moreover, because a real robot was used, the robot's boarding speed was limited to its range of speed. It is also difficult to ensure the safety of the participants when a robot moves at high speeds, making it challenging to evaluate passenger impressions in an elevator car using an actual robot.

On the other hand, Future boarding behavior of robots should be considered as an extension of current robots. Just as humans have eyes only on the front side of their bodies, most current service robots have sensors such as Lidar only on the front of their own. When service robots move forward and get into elevator cars, they need to rotate 180 degrees within the car to prepare for their exit. These sequences of boarding behavior should be expected to be carried over to future service robots.

In this study, we aimed to enable the advance evaluation of the impressions of future high-performance robots and developed a mixed reality (MR) system that incorporates a virtual mobile robot into the real world (Fig. 1). We used this system and investigated the impressions and stress levels that passengers experience due to the boarding behavior of the virtual mobile robot with human-like high mobility.

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Figure 1. Virtual robot riding in a real elevator using a mixed reality system

II. MIXED REALITY SYSTEM

We developed an MR system that incorporates a virtual robot into the real-world with a head-mounted display (HMD; Meta Quest 3 was used) (Fig. 2). For the image of a virtual robot, the 3D data of a mobile robot was used [16] (a 1420-mm-tall, 500-mm-wide, and 630-mm-deep mobile robot with a shelf on a mobile base). The initial position of the virtual robot was determined with respect to the HMD coordinate system. The virtual robot was displayed with the bottom touching the floor using spatial information obtained from the HMD. When there was an occlusion such as elevator doors between the virtual robot and the HMD wearer, the virtual robot was set to not be displayed.

When the left trigger of the HMD controller was pulled, the position of the virtual robot was reset. When the right trigger of the HMD controller was pulled, the virtual robot was programmed to perform a series of movements: moving forward a specified distance at a specified speed, rotating 180 degrees at a specified rotation speed, and then stopping. It was possible to change the forward movement speed (high/low), rotation speed (high/low), and rotation direction (clockwise (CW)/counterclockwise (CCW)) of the virtual robot using HMD controllers.

The low forward and rotation speeds were determined as the forward and rotation speeds of the original robot (0.3 m/s and 40 deg/s, respectively). The high forward and rotation speeds were determined based on typical human mobility (1.5 m/s and 360 deg/s, respectively). To enhance the immersive experience of mixed reality, motion sounds based on the actual robot were played while the virtual robot was in motion. Specifically, the forward sound and rotation sound of the virtual robot at low speeds were recorded from the actual robot moving forward and rotating at the specified speeds, respectively. The forward sound and rotation sound of the virtual robot at high speeds were modified versions of the recorded sounds of the actual robot moving forward and rotating at its maximum speeds, respectively.

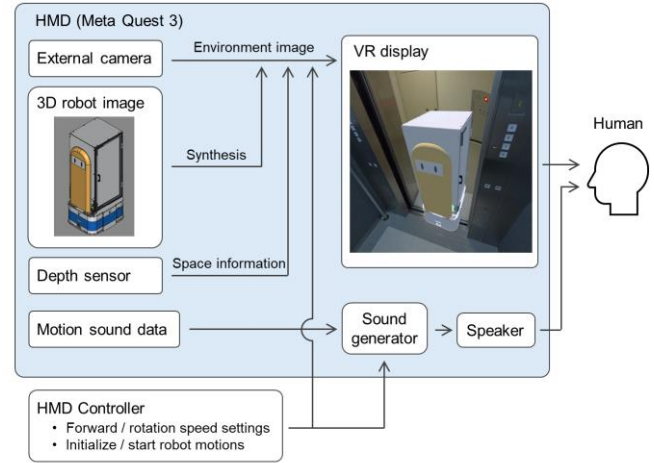


Figure 2. Developed mixed reality system

III. EXPERIMENT

A. Hypotheses and prediction

As mentioned in the previous section, the boarding behavior of current mobile robots can be classified into two basic movements: moving forward to enter the elevator car and rotating 180 degrees inside the car. When a passenger boards the elevator with a mobile robot, these two movements can be further divided into three factors: forward speed, rotation speed, and rotation direction (facing toward or backing away from the passenger).

As an extension of the previous research [15], it is easily expected that quick robot boarding behavior like humans will impart a more positive impression to passengers inside the elevator. On the other hand, since the robot speed in [15] was slower compared to humans, passengers may perceive robot movements at human-like speeds as too fast, which may give a negative impression on them. Therefore, we formulated the following two opposing hypotheses:

H1: When the virtual robot boards an elevator at a human-like high forward and rotation speeds, participants will have more positive impressions of the robot than when it boards at a low speed.

H0: When the virtual robot boards an elevator at a human-like high forward and rotation speeds, participants will have more negative impressions of the robot than when it boards at a low speed.

Mobile robots are currently programmed to turn in a predetermined direction inside elevator cars, regardless of whether a passenger is present or not. However, when a mobile robot has a face-like interface located on its front, as the robot in this study did, that interface will act like a human face. We hypothesized that when the robot's interface turns to face the passenger, it would give the passenger an impression similar to that of a person turning their face toward them. Following this hypothesis, we also prepared the following prediction:

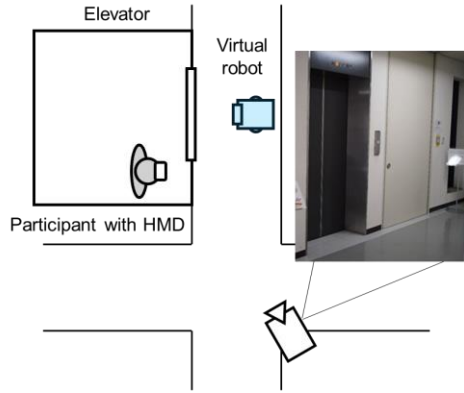


Figure 3. Experimental environment

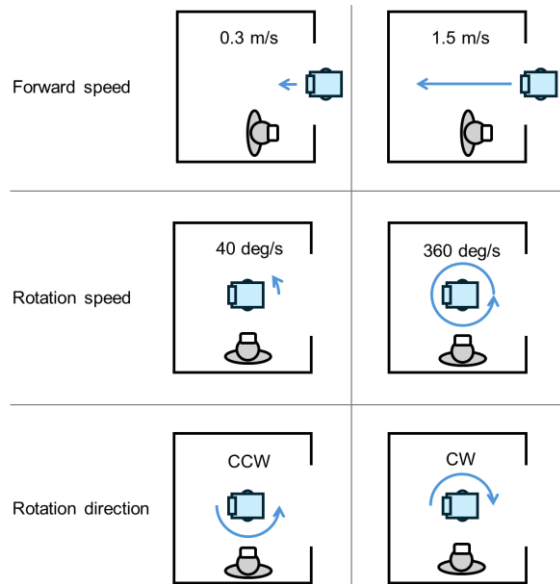


Figure 4. Illustration of experiment factors

Prediction: When the virtual robot turns to face participants in the elevator, their likeability increases, but their stress levels also increase.

B. Participants

Thirty people participated in the experiment: 15 women and 15 men. Their ages ranged from 20s to 50s, and their average age was 39.7 (S. D. = 10.6). They applied for the experiment through a temporary employment agency.

C. Environment

Figure 3 shows the experimental environment. We used our MR system explained in the last section and an elevator in our laboratory. The elevator had operation panels on both sides of the door (Fig. 1). The size of the elevator was 2.3 m high, 1.35 m wide, and 1.6 m deep. We installed a network camera inside the elevator car to check on participants.

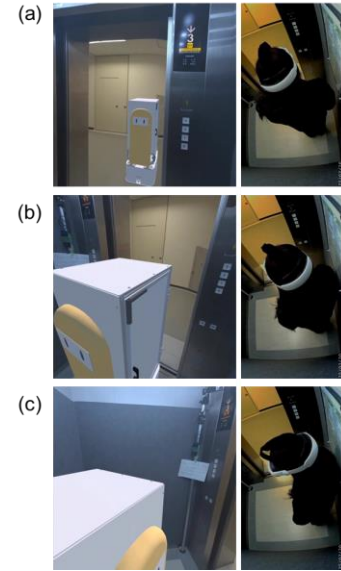


Figure 5. Sequence of the experiment. Left shows a participant's view and right shows corresponding objective view from the wireless camera inside the elevator. (a) Participant waits virtual robot in elevator. (b) Virtual robot enter the elevator. (c) Virtual robot rotates 180 degrees and then stops.

D. Condition

We considered three factors (Fig. 4): *forward speed* (0.3 m/s or 1.5 m/s), *rotation speed* (40 deg/s or 360 deg/s), and *rotation direction* (CCW or CW). We thus prepared eight conditions. We fixed the standing position of participants in the elevator car as shown in Figure 3. When rotation direction is CCW, the virtual robot turned so that it faced toward the participants. When rotation direction is CW, the virtual robot turned so that it faced away from the participants.

E. Measurements

In this study, we evaluated the perceived impressions of the virtual robot with existing questionnaire scales [17]: anthropomorphism, animacy, likeability, perceived intelligence, and safety. The items were assessed using a 1-to-7 response format, where 1 denoted the most negative answer and 7 denoted the most positive. We also evaluated the perceived stress level using a 0-to-10 response format, with 0 indicating no stress and 10 indicating the most stressful.

F. Procedure

All the procedures were approved by the Advanced Telecommunication Research Review Boards (525H). First, the participants read explanations about the experiment and how to evaluate the virtual robot in each condition. We employed a within-participant design where the participants experienced eight different conditions. After starting the experiment, the participants first got on the elevator and waited for the virtual robot to board (Fig. 5a). The virtual robot then entered the elevator car with the specified forward speed (Fig. 5b), stopped at the center of the car, and rotated 180 degrees with the specified rotation speed (Fig. 5c). After the virtual robot stopped inside the car, the participants got off the elevator

and answered questionnaires. The order of the conditions was randomized. At the end of the experiments, we conducted a brief interview with the participants.

IV. RESULTS

A. Questionnaire results

We conducted a three-factor (*forward speed*, *rotation speed*, and *rotation direction*) ANOVA to analyze the questionnaire results (Figs. 6-11). The statistical analysis of the virtual robot's anthropomorphism scale showed a significant difference in the *forward speed* factor ($F(1,29) = 10.58$, $p = 0.003$, $\text{partial } \eta^2 = 0.267$), and no significant differences in the *rotation speed* factor, in the *rotation direction* factor, or in the interaction effects.

The statistical analysis of the virtual robot's animacy scale showed no significant differences in the main and interaction effects of all factors.

The statistical analysis of the virtual robot's likeability scale showed a significant difference in the *forward speed* factor ($F(1,29) = 27.93$, $p < 0.001$, $\text{partial } \eta^2 = 0.491$), and the *rotation speed* factor ($F(1,29) = 5.798$, $p = 0.023$, $\text{partial } \eta^2 = 0.167$), and no significant differences in the *rotation direction* factor or in the interaction effects.

The statistical analysis of the virtual robot's intelligence scale showed a significant difference in the *forward speed* factor ($F(1,29) = 6.497$, $p = 0.016$, $\text{partial } \eta^2 = 0.183$), and no significant differences in the *rotation speed* factor, in the *rotation direction* factor, or in the interaction effects.

The statistical analysis of the virtual robot's safety scale showed a significant difference in the forward speed factor ($F(1,29) = 32.72$, $p < 0.001$, $\text{partial } \eta^2 = 0.530$), and no significant differences in the *rotation speed* factor, in the *rotation direction* factor, or in the interaction effects.

Finally, the statistical analysis of the stress level scale showed a significant difference in the *forward speed* factor ($F(1,29) = 29.12$, $p < 0.001$, $\text{partial } \eta^2 = 0.501$), and no significant differences in the *rotation speed* factor, in the *rotation direction* factor, or in the interaction effects.

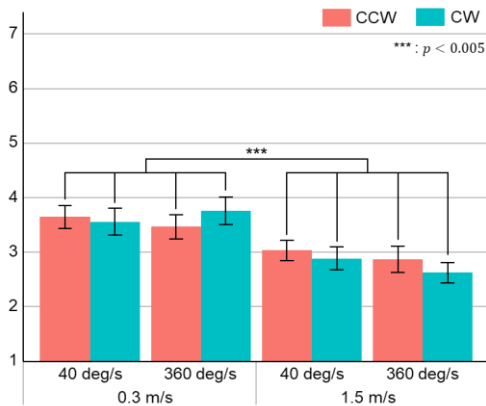


Figure 6. Questionnaire results of virtual robot's anthropomorphism (average and S.E.)

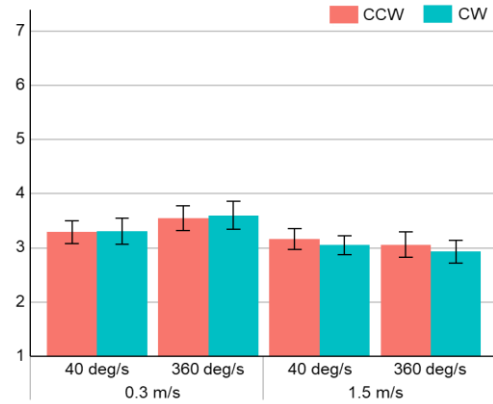


Figure 7. Questionnaire results of virtual robot's animacy (average and S.E.)

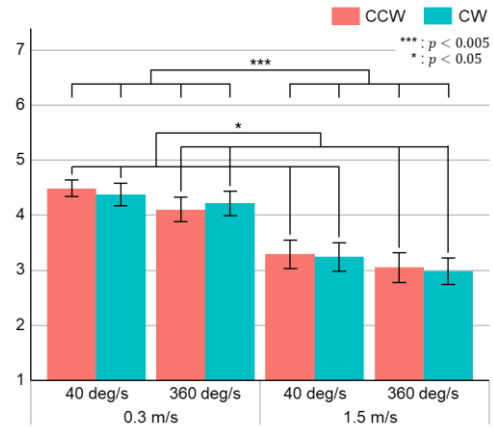


Figure 8. Questionnaire results of virtual robot's likeability (average and S.E.)

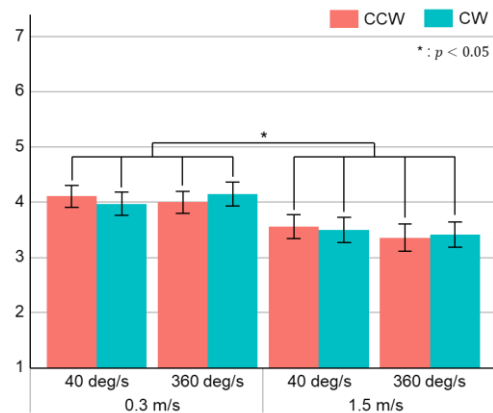


Figure 9. Questionnaire results of robot's intelligence (average and S.E.)

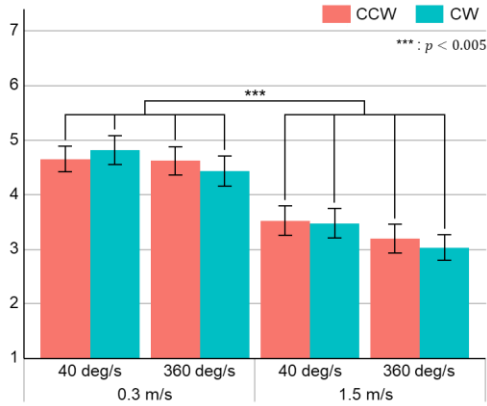


Figure 10. Questionnaire results of virtual robot's safety (average and S.E.)

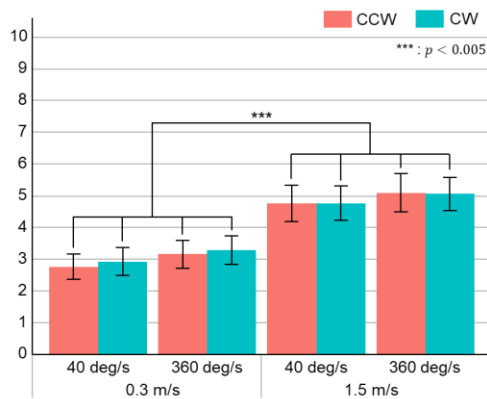


Figure 11. Questionnaire results of perceived stress (average and S.E.)

B. Summary

H0 was partially supported. The participants had negative impressions of the virtual robot when it rode the elevator at human-like forward speed in the context of anthropomorphism, likeability, intelligence, safety, and stress levels. The participants also had negative impressions of the virtual robot when it rotated inside the elevator at high rotation speed in the context of likeability. Our prediction was not able to be verified. There was no statistically significant difference in rotation direction.

V. DISCUSSIONS

A. Implication

In recent years, MR systems have been developed for robot operations and controls [18-21]. In this study, we constructed an MR system to investigate the impressions that human-like, high-speed elevator boarding by robots gives to the passengers in the elevator car. We developed the system to ensure participants' safety as it never physically affects them. Despite that, most participants expressed fear or surprise in their post-experiment interviews, especially about the virtual robot's high-speed boardings. We found that our mixed reality system can evoke emotions in participants similar to those

experienced when a real mobile robot is present. This will be useful when designing future interactions in confined spaces, such as inside an elevator.

Our results show that riding an elevator at high forward speed, similar to human walking speeds, decreases perceived anthropomorphism, likeability, intelligence, and safety, and increases stress levels in a statistically significant manner. In addition, our results also show that rotating inside an elevator at high rotation speed decreases perceived likeability in a statistically significant manner. These partly support H0, indicating that too fast robot ride and rotation impart more negative impressions to passengers than robot ride and rotation at normal speeds. It is interesting that although our virtual robot's high-speed ride (1.5 m/s) falls within the range of normal human walking speeds [22], participants expressed fear or surprise at the robot's high-speed forward entry. This suggests that there is an appropriate motion speed range for mobile robots in confined spaces. It is also possible that participant fear or surprise when the virtual robot approaches at high speed outweighed the reduction in stress from the elimination of waiting time for the robot to board the elevator.

Our results also showed that the difference in rotation speed of the virtual robot was not statistically significant, except for likeability. It seems interesting because almost all subjects indicated that they preferred the slower rotation speed of the virtual robot in the post-experiment interviews. One possible reason is that the approach of the virtual robot left a strong impression on the participants, similar to that of a real robot [23-25], which overshadowed their impressions of the robot rotation. In addition, we were unable to verify our prediction because no statistically significant difference was found for the direction of the robot's rotation. We might verify the prediction if the shape of the virtual robot was closer to that of a human, as was the case in some previous studies [25,26]. Our results suggest that rotation speed partially affects passenger impressions, while rotation direction has no significant effect on passenger impressions. This finding would serve as a guideline for the hardware and control designs of mobile robots.

B. Limitation

There are some limitations that should be noted. For example, it is necessary to evaluate our MR system by comparing the perceived impressions of the virtual robot with those of the original robot. In this study, although we provided fundamental insights for future experiments, the difference between the high and low speeds of forward and rotational motions was quite significant. Hence, it may be necessary to investigate intermediate speeds in future studies. From an engineering standpoint, the appropriate motion speed range for mobile robots and the effects of robots' shape and size on passenger impressions should also be investigated. We would like to address these issues in future research.

VI. CONCLUSION

In this study, we developed an MR system to evaluate quick, human-like boarding behavior that a mobile robot will

demonstrate in the future. The MR system introduced a virtual robot into the real world, providing visual and auditory information about the robot through an HMD. For the experiment using the MR system, we decomposed robot's boarding behavior into three basic factors. We then investigated the impressions and stress level of passengers with respect to the virtual robot's boarding behavior based on those factors. A statistical analysis revealed that the virtual robot riding on an elevator at a human-like forward speed gave more negative impressions and caused more stress when compared to normal robot riding speed. The statistical analysis also showed that the virtual robot rotation inside an elevator at a high rotation speed gave a more negative impression when compared to normal robot rotating speed. On the other hand, the statistical analysis showed no significant differences in the impressions or in the stress levels in relation to the robot's rotation direction. These results provide useful knowledge for designing the behavior of future robots when they board elevators and for designing human-robot interactions in confined spaces.

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